

Tsunami lessons

The world **breathed** easy as Japan and Hawaii **downgraded**, and Russia **lifted** its tsunami warnings. No **casualties** were reported till late evening although damage was **extensive** in Russia's **coastal** town of Severo-Kurilsk, **battered** by waves 10-13ft high. The town's 2,000-odd population lives at a higher ground and is safe. The world was **on tenterhooks** because tsunamis **spell devastation** – almost 2.5L died in 2004, over 20,000 in Japan in 2011. One reason casualties are **so far** nil, and damage limited, is science, **forecasting** and international **collaboration**. Scientists can **predict** not only when a tsunami will hit but also how much flooding there'll be, **unthinkable** in 2004. Both the tech and coverage of Pacific Ocean – where over 75% of tsunamis form – with sensors, early warning **systems** and deep-ocean **assessment** (US NOAA's DART **buoys**) **have** improved many times the last two **decades**.



Ocean **quakes** **trigger** waves on the ocean floor – the difference between a tsunami wave and a surface one. These push out with force that sensors **deep** in the Pacific now record via sonics and other indicators. If in 2004, there were only six such buoys, there are over 40 now **scattered** across **seismically** active regions in Pacific Ocean. What **makes a difference** in impact on **coastlines** is the **incline** from ocean floor to the coast. Sharper the incline, higher the body of water pushed up, bigger the wave. This makes Japan among the most **vulnerable** because its incline is **steep**. But Japan also has among the most sophisticated warning systems. An earthquake still can't be **predicted** but its **epicentre** can be quickly calculated and warnings issued within seconds and minutes through **seismographs** nationwide. Authorities have time to respond and **swiftly evacuate** people in the line of a **brute** wall of water. There's no being pacific about that.

[Practice Exercise]

- Red/blue coloring of words in the sentence indicate subject verb relationship; where 'red' denotes 'subject' and 'blue' denotes 'verb'.
- **Pacific** (adjective) – peaceful in character or intent. शान्तिप्रद

Vocabulary

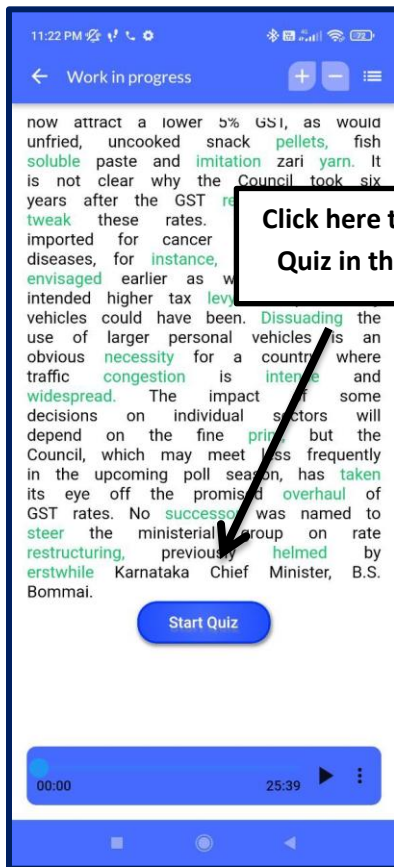
1. **Breathe** (verb) – inhale, exhale, respire, gasp, puff साँस लेना
2. **Downgrade** (verb) – reduce, lower, demote, devalue, diminish घटाना
3. **Lift** (verb) – remove, revoke, withdraw, end, eliminate हटाना
4. **Casualty** (noun) – victim, fatality, injured, sufferer, loss हताहत
5. **Extensive** (adjective) – widespread, broad, vast, massive, comprehensive व्यापक
6. **Coastal** (adjective) – seaside, shoreline, littoral, maritime, beachfront तटीय
7. **Batter** (verb) – strike, pound, hit, smash, beat क्षतिग्रस्त करना
8. **On tenterhooks** (phrase) – anxious, uneasy, nervous, worried, apprehensive बेचैन
9. **Spell** (noun) – period, stretch, bout, phase, interval अवधि
10. **Devastation** (noun) – destruction, ruin, havoc, demolition, annihilation तबाही
11. **So far** (phrase) – until now, thus far, up to this point, as yet, till date अब तक
12. **Forecast** (verb) – predict, project, foresee, estimate, anticipate पूर्वानुमान लगाना
13. **Collaboration** (noun) – cooperation, partnership, teamwork, alliance, association सहयोग
14. **Predict** (verb) – forecast, foresee, prophesy, anticipate, estimate पूर्वानुमान लगाना
15. **Unthinkable** (adjective) – unimaginable, inconceivable, incredible, unbelievable, absurd अकल्पनीय
16. **Assessment** (noun) – evaluation, appraisal, estimation, judgment, analysis मूल्यांकन
17. **Buoy** (verb) – float, support, uplift, hold up, stabilize तैराना
18. **Decade** (noun) – Period of ten years दशक
19. **Quake** (noun) – earthquake, tremor, shock, jolt, seismic activity भूकंप
20. **Trigger** (verb) – cause, activate, initiate, spark, provoke ट्रिगर करना
21. **Deep** (adjective) – profound, bottomless, abyssal, extensive, immersed गहरा
22. **Scatter** (verb) – disperse, spread, strew, distribute, sprinkle बिखेरना
23. **Seismically** (adverb) – tectonically, geologically, vibrationally, earth-shakingly, fault-based भूकंपीय रूप से

24. **Make a difference** (phrase) – have an impact, affect, influence, matter, count
फर्क डालना
25. **Coastline** (noun) – shore, seaboard, beach, coastal edge, waterfront समुद्र तट
26. **Incline** (noun) – slope, gradient, rise, tilt, pitch ढलान
27. **The vulnerable** (noun) – the weak, the exposed, the defenseless, the at-risk, the unprotected असुरक्षित लोग
28. **Steep** (adjective) – sharp, abrupt, vertical, inclined, sheer खड़ी
29. **Predict** (verb) – forecast, foresee, project, anticipate, estimate भविष्यवाणी करना
30. **Epicentre** (noun) – focus, hypocenter, origin, nucleus, starting point उपकेंद्र
31. **Seismograph** (noun) – quake recorder, tremor detector, ground-motion sensor, vibration meter, seismic device भूकंपमापक
32. **Swiftly** (adverb) – quickly, rapidly, speedily, promptly, immediately शीघ्रता से
33. **Evacuate** (verb) – vacate, empty, abandon, clear out, relocate हटा देना, खाली करना
34. **Brute wall of water** (phrase) – a powerful and overwhelming force of water, often appearing like an impassable barrier.

Summary of the Editorial

1. **Downgraded Alerts:** Japan and Hawaii downgraded tsunami alerts; Russia lifted its warning, easing global anxiety.
2. **No Casualties Reported:** Despite extensive damage in Russia's Severo-Kurilsk, no deaths were reported due to effective precautions.
3. **Past Disasters Remembered:** The fear of tsunamis stems from past devastation — 2.5 lakh dead in 2004 and over 20,000 in Japan in 2011.
4. **Scientific Advancements:** Casualties were avoided this time largely due to scientific progress in tsunami forecasting.
5. **Improved Prediction Models:** Today, scientists can estimate not just the time of tsunami landfall but also potential flood levels.
6. **Tech-Enhanced Monitoring:** There have been significant improvements in early warning systems and ocean monitoring technology.
7. **DART Buoys Deployment:** From just 6 in 2004, now over 40 NOAA DART (Deep-ocean Assessment and Reporting of Tsunamis) buoys monitor seismic activity in the Pacific.
8. **Tsunami Origins Explained:** Unlike surface waves, tsunami waves are triggered by ocean-floor quakes, generating immense energy.
9. **Sensor Detection:** Modern sensors use sonics and other indicators to detect deep-sea seismic movements early.
10. **Impact of Coastal Incline:** The angle of seafloor rise affects wave height — a steep incline produces higher and more destructive waves.
11. **Japan's High Vulnerability:** Japan is especially vulnerable due to its steep ocean floor rise close to the coast.
12. **Sophisticated Japanese Systems:** Japan compensates for its risk with highly advanced warning systems and national seismograph networks.
13. **Rapid Response Capabilities:** Japan can calculate earthquake epicentres within seconds and issue timely alerts for evacuation.
14. **Effective Evacuation Saves Lives:** Swift evacuation plans and public readiness significantly reduce casualties.
15. **Global Collaboration Key:** International scientific cooperation and shared data play a vital role in disaster preparedness and response.

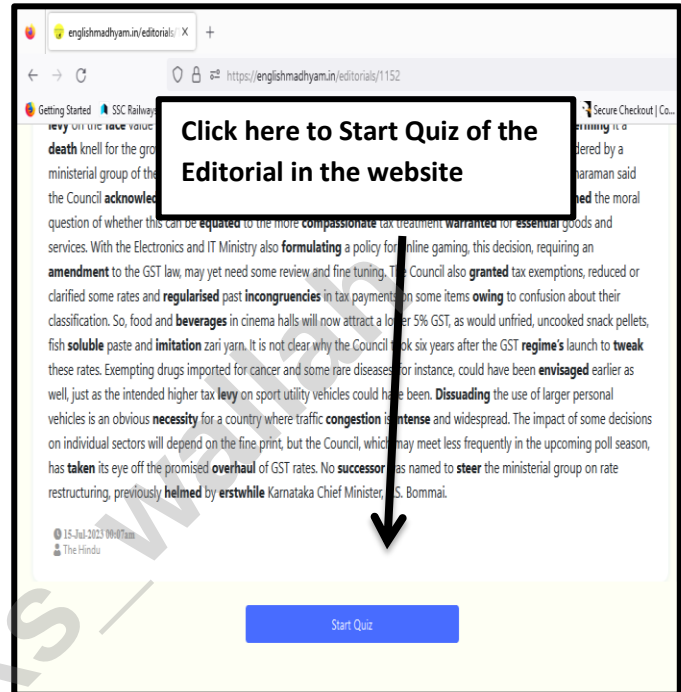
Practice Exercise: SSC Pattern Based



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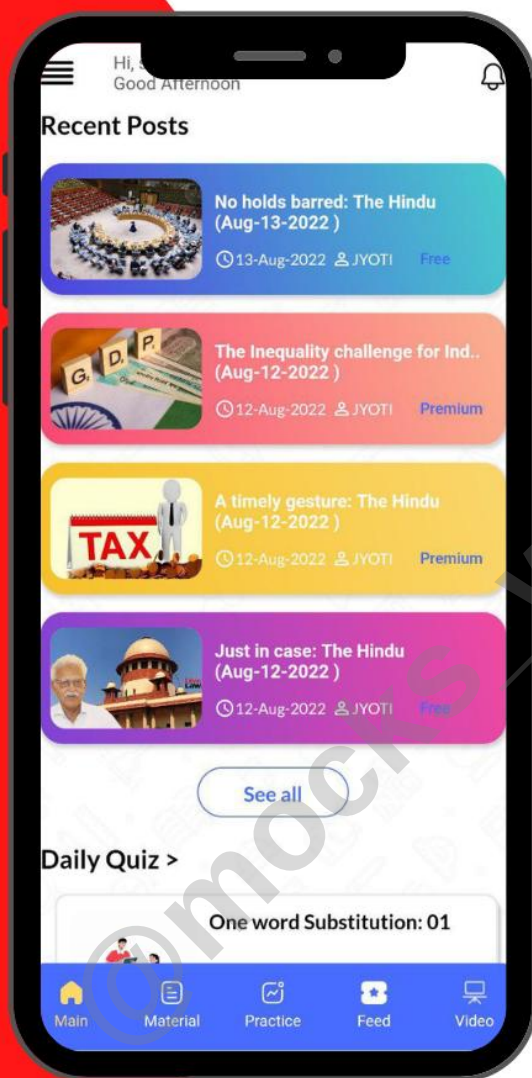
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